

Project Proposal: Multiplayer Game

Design Architecture & Specifications

PROJECT NAME	Robot Rescue Team (Multiplayer Edition)	DEVELOPMENT TIMELINE	2 Weeks
TARGET AUDIENCE	Early Childhood to Early Primary (FS1 to Grade 3)	CORE OBJECTIVE	Teach logic, planning, and problem-solving through a competitive team-vs-team maze race (1v1 or 2v2) across different devices and WiFi networks.

Design Option: The "Command Queue" Race

Strategy-Focused Framework

Executive Summary

This design preserves the original educational framework of the game. Instead of controlling characters in real-time, students must plan their path, build a sequence of instructions, and test their logic against the opposing team.

How the Gameplay Works

- **The Planning Phase:** Players look at the maze layout. They click on-screen buttons to build a visible list of steps (e.g., **Move Forward** → **Turn Left** → **Pick Object**).
- **The Execution Trigger:** Once the team agrees on the sequence, they press the **Run** button. The robot executes the entire list of instructions automatically while the players watch.
- **Handling Mistakes:** If the robot hits a wall or misses an item due to a logic error, it stops. The team presses **Reset**, the robot returns to the starting tile, and they edit their instruction list to try again.
- **Winning Condition:** The first team whose robot successfully completes the mission objectives and reaches the goal wins the round.

Technical Impact (Unity Netcode + Unity Gaming Services)

- **Network Traffic:** Very Low. Devices only exchange short strings of text data (the instruction lists), requiring minimal internet speed.
- **Lag Tolerance:** Excellent. Slow WiFi or high latency will not give any team an unfair advantage, as all command processing happens cleanly on the local device.

Feature Specifications Matrix

Feature	Option A: Command Queue
Control Input & Pace of Play	Build a command list first, then press Run. Deliberate, strategic, and analytical.
Core Skill Taught	Algorithmic sequencing and debugging.
WiFi Requirements	Highly forgiving of slow internet or lag spikes.
UI Design Complexity	Higher (Requires data list displays and editing tools).